

Q1: a:

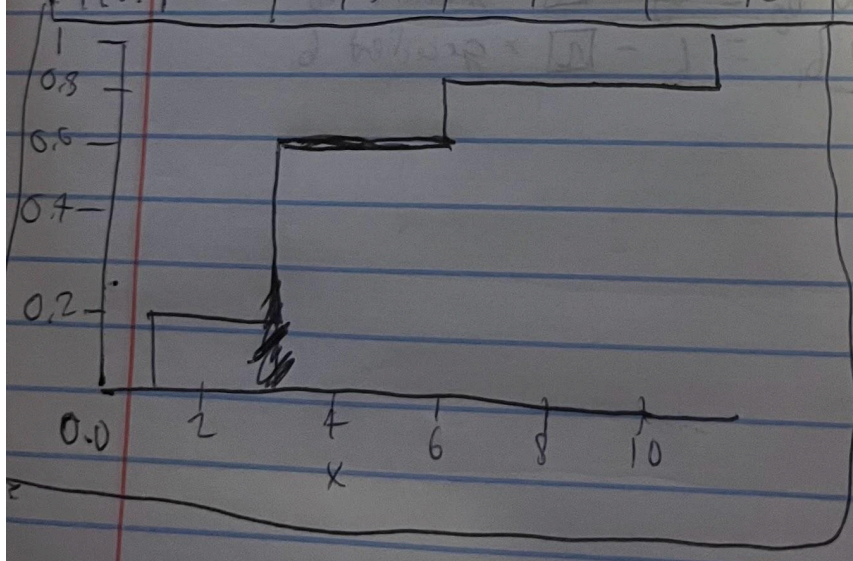
obs	red	blue	green	grey
red	1	0	0	0
blue	0	1	0	0
blue	0	1	0	0
green	0	0	1	0
grey	0	0	0	1
red	1	0	0	0

b: Samp. prop. of each column is the mean of its column:

$$\hat{p}_{red} = \frac{2}{6} = \frac{1}{3}, \hat{p}_{blue} = \frac{2}{6} = \frac{1}{3}, \hat{p}_{grey} = \frac{1}{6}, \hat{p}_{green} = \frac{1}{6}$$

c: ECDF is $\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n I(x_i \leq x)$ → proportion of the sample at or below x . With $n=5$, steps up by $1/5$ at each observation, and by $2/5$ at 3 (6/5 3 appears twice).

$x < 1$	$1 \leq x < 3$	$3 \leq x < 6$	$6 \leq x < 11$	$x \geq 11$	
$F(x)$	0	$1/5$	$3/5$	$4/5$	1



Q2:

a. Since $b > 0$, event $Y \leq y$ is the same as $a + bX \leq y$, which is $X \leq \frac{y-a}{b}$, so: $F_Y(y) = P[Y \leq y] = P[X \leq \frac{y-a}{b}] = F_X(\frac{y-a}{b})$

and density is: $f_Y(y) = \frac{d}{dy} F_X(\frac{y-a}{b}) = \frac{1}{b} f_X(\frac{y-a}{b})$

b. $\bar{X} = \frac{1+3+4}{3} = \frac{8}{3} \approx 2.67$

c. $X^2 = [1, 9, 16]$ & $\overline{X^2} = \frac{1+9+16}{3} = \frac{26}{3} \approx 8.67$

d. $\overline{X^2} = (\frac{8}{3})^2 = \frac{64}{9} \approx 7.11$

This is not the same as part c, the mean of the squares (8.67) is larger than the square of the mean, and the difference b/t them, 1.56, is the sample variance.

Q3:

a: $F(x)$;

CDF

1
0.8
0.6
0.4
0.2
0.0

0 2 3 4 5 6

$f(x)$;

PDF

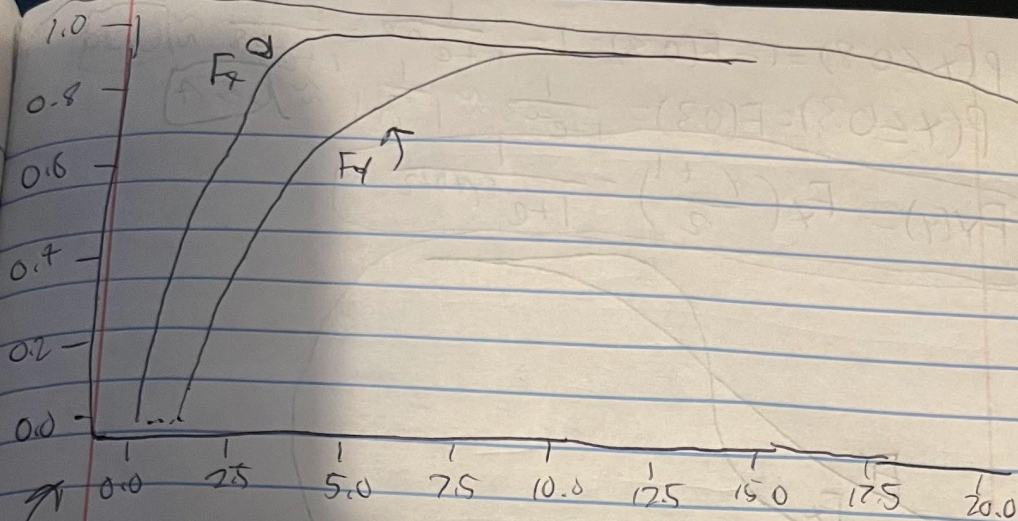
1.0
0.8
0.6
0.4
0.2
0.0

1 2 3 4 5 6

$$b: P(X \geq 3) = 1 - F(3) = 1 - (1 - e^{-3}) = e^{-3} \approx 0.050$$

$$P(4 \leq 35) = F(35) = 1 - e^{-35} \approx 1$$

$$c: F_Y(y) = F_X\left(\frac{y-1}{3}\right) = \begin{cases} 0, & y < 1 \\ 1 - e^{-(y-1)/3}, & y \geq 1 \end{cases}$$

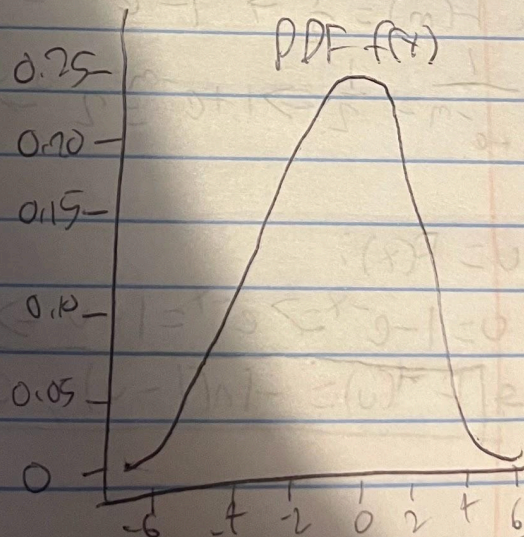
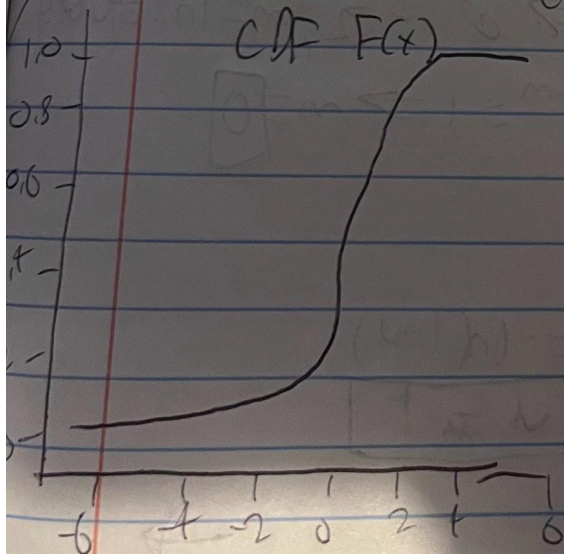


(Compared w/ F_X , the curve F_Y starts at 1 and rises three times more slowly.

$$P(Y < 10) = F_Y(10) = 1 - e^{-(10-1)/3} = 1 - e^{-3} \approx 0.95$$

Q4:

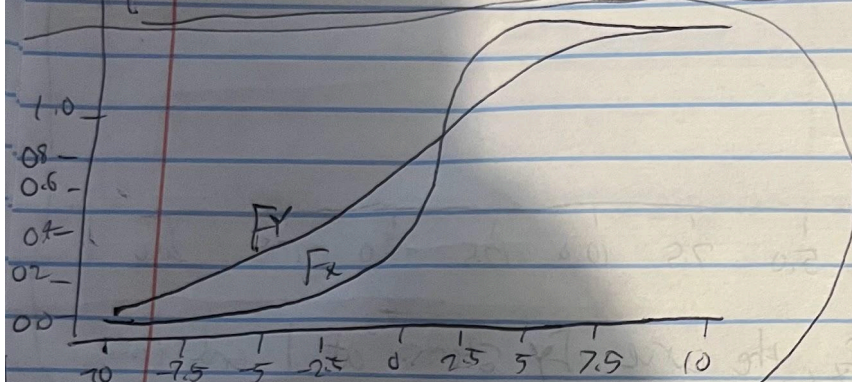
$$a) f(x) = F'(x) = \frac{d}{dx}(1 + e^{-x})^{-1} = \frac{e^{-x}}{(1 + e^{-x})^2}$$



$$b: P(X \geq 0.8) = 1 - F(0.8) = 1 - \frac{1}{1 + e^{-0.8}} = \frac{e^{-0.8}}{1 + e^{-0.8}} \approx \boxed{0.310}$$

$$P(X \leq 0.3) = F(0.3) = \frac{1}{1 + e^{-0.3}} \approx \frac{1}{1.741} \approx \boxed{0.574}$$

$$c: F_Y(y) = F_X\left(\frac{y+1}{2}\right) = \frac{1}{1 + e^{-(y+1)/2}}$$



F_Y rises half as steeply.

$$P(Y < 0) = F_Y(0) = \frac{1}{1 + e^{-1/2}} \approx \frac{1}{1.607} \approx 0.622$$

25:

$$a: F(m) = \frac{1}{2} \Rightarrow 1 - e^{-m} = \frac{1}{2} \Rightarrow e^{-m} = \frac{1}{2}, m = \ln 2 \approx \boxed{0.693}$$

$$b: \frac{1}{1 + e^{-m}} = \frac{1}{2} \Rightarrow 1 + e^{-m} = 2 \Rightarrow e^{-m} = 1 \Rightarrow m = \boxed{0}$$

c: $u = F(x)$:

$$u = 1 - e^{-x} \Rightarrow e^{-x} = 1 - u \Rightarrow x = -\ln(1 - u)$$

$$\Rightarrow \boxed{F^{-1}(u) = -\ln(1 - u), 0 \leq u \leq 1}$$

$$\delta. u = \frac{1}{1+e^x} \Rightarrow 1+e^x = \frac{1}{u} \Rightarrow e^x = \frac{1-u}{u}$$

$$\Rightarrow x = \ln\left(\frac{u}{1-u}\right) \Rightarrow \boxed{F^{-1}(u) = \ln\left(\frac{u}{1-u}\right), 0 < u < 1}$$

QUESTION 6:

Q6.

Load `./data/metabric.csv`.

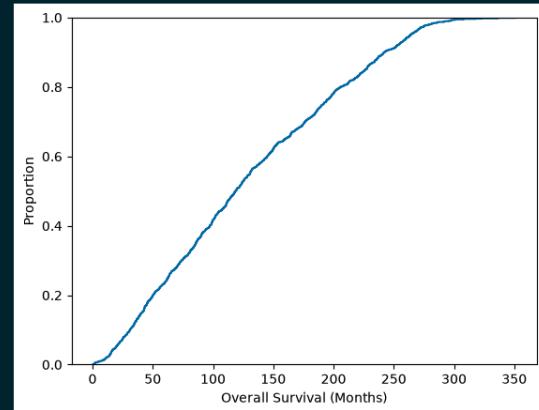
- a. Make an ECDF plot of `Overall Survival (Months)`.
- b. Make an ECDF plot of `Overall Survival (Months)`, hued by `Chemotherapy`. Conditional on a patient receiving Chemotherapy, should we predict they will live a longer or shorter amount of time? Explain your answer clearly.
- c. Is chemotherapy an effective treatment? Explain your answer clearly.

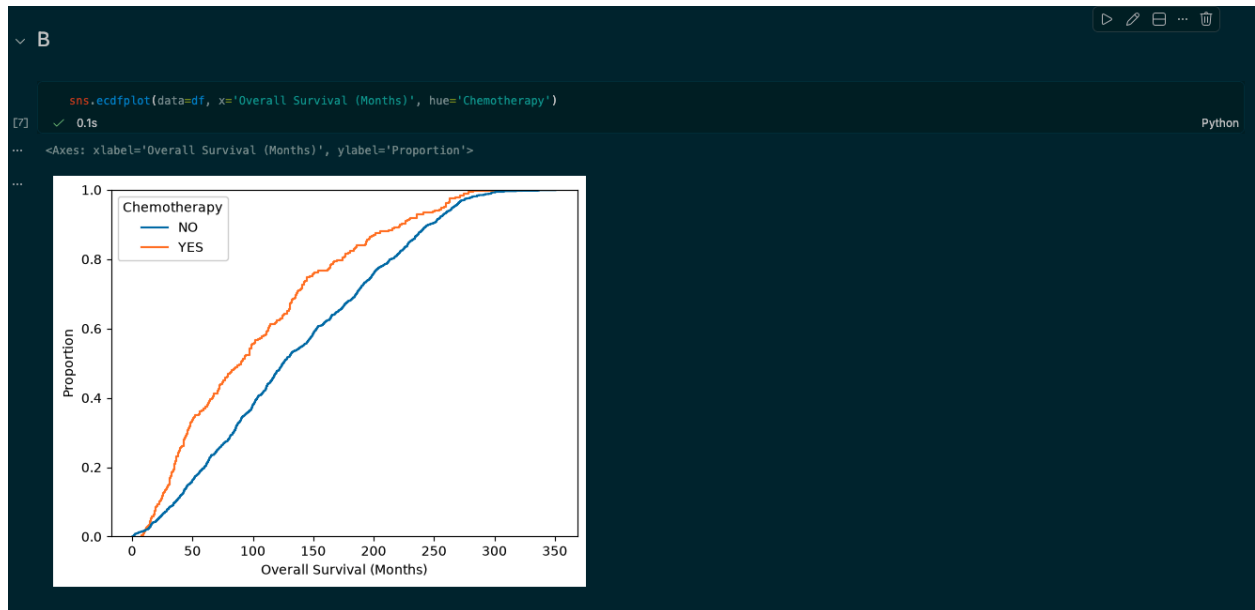
```
import pandas as pd
import seaborn as sns
df = pd.read_csv('./data/metabric.csv')

sns.ecdfplot(data=df, x='Overall Survival (Months)')
```

[6] ✓ 0.2s

Python





Looking at the above graphs, we can see that the chemotherapy ecdf sits above and to the left of the no-chemo ecdf at every survival time, and at any number of months x , a larger proportion of chemo patients had already died by x . Indeed, if we read off the curves, we can see that at a proportion of 0.5, the median survival is around 90 months with chemo and around 125 months without chemo. Thus, conditional on a patient receiving chemo, we should predict that they will live a shorter amount of time.

PART C:

We cannot conclude from this data that chemo is effective or ineffective. Indeed, part c is a causal question, and asks does chemo make a patient live longer than they otherwise would? Just looking at the ecdf cannot answer the second question as patients were not randomly assigned to chemo. Indeed, doctors are likely to give chemo to patients with more severe levels of cancer, so the chemo group was expected to live a shorter amount of time regardless of whether or not they were treated, and thus their shorter survival reflects who gets the treatment. In order to judge the effectiveness, we would need to compare chemo and no-chemo patients who are otherwise alike, ideally from a randomized trial.